

TOLCHESTER BEACH, MD, USA

WMO: 997312

Lat: 39.2135N

Lon: 76.2446W

Elev: 10

StdP: 101.20

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.8	(c) -6.4	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 14.9	(k) -2.7	(l) 13.7	(m) -2.5	(n) 6.7	(o) 310	(p) 0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.8	(c) 31.8	(d) N/A	(e) 30.7	(f) N/A	(g) 29.6	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 2.6	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 13.4	(b) 11.4	(c) 9.7	(d) DB	(e) -10.6	(f) 34.0	(g) 2.5	(h) 1.5	(i) -12.4	(j) 35.1	(k) -13.9	(l) 36.0	(m) -15.3	(n) 36.8	(o) -17.0	(p) 37.9	(q) N/A
(4)			(r) WB	(s) N/A	(t) N/A	(u) N/A	(v) N/A	(w) N/A	(x) N/A	(y) N/A	(z) N/A	(aa) N/A	(ab) N/A	(ac) N/A	(ad) N/A	(ae) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	1.4	2.3	6.6	13.3	18.6	23.4	26.0	24.8	21.3	15.3	9.1	4.8
	DBStd	9.38	4.76	4.56	4.57	4.24	3.70	2.92	2.28	2.38	3.18	3.96	4.00	4.22
	HDD10.0	867	268	218	126	15	0	0	0	0	5	64	171	
	HDD18.3	2357	524	448	363	161	43	3	0	0	8	109	278	419
	CDD10.0	2315	2	4	21	113	268	403	495	457	338	169	36	9
	CDD18.3	766	0	0	0	9	52	156	237	199	97	16	0	0
	CDH23.3	5502	0	0	1	52	225	1031	2135	1521	495	43	0	0
	CDH26.7	1395	0	0	0	9	34	227	649	379	91	6	0	0
(14) Wind	WSAvg	3.6	4.3	4.2	4.4	4.1	3.4	3.2	2.8	2.6	2.8	3.6	4.1	4.1
(15) Precipitation	PrecAvg	1141	90	68	110	92	93	100	109	97	118	95	78	90
	PrecMax	1643	156	178	219	140	183	307	203	195	411	213	147	208
	PrecMin	837	34	7	0	43	22	33	43	18	24	5	24	31
	PrecStd	186	31	37	48	30	46	63	45	42	80	64	40	47
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.9	18.2	21.7	28.0	29.8	32.2	34.0	32.9	30.9	27.0	20.9	18.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	13.7	14.2	18.5	24.7	27.5	30.4	32.3	31.3	29.2	24.4	18.8	16.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	10.8	10.8	15.8	22.1	25.8	29.1	31.0	30.0	27.5	22.7	16.8	13.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	7.8	8.6	13.6	19.7	24.3	27.9	30.0	28.8	26.2	21.2	15.2	11.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	6.5	7.1	8.1	7.6	6.8	6.8	6.8	7.1	6.7	6.7	5.9
		MCDBR	11.1	10.0	11.4	11.2	8.6	7.7	7.5	7.4	8.0	7.8	8.8	8.9
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.312	0.320	0.366	0.400	0.447	0.499	0.518	0.513	0.426	0.363	0.327	0.310
	taud		2.469	2.435	2.359	2.290	2.194	2.092	2.052	2.051	2.269	2.451	2.508	2.517
	Ebn at Noon		861	904	889	877	838	793	772	764	816	843	837	838
	Edh at Noon		80	94	113	128	144	160	165	162	122	92	76	71
(44) All-Sky Solar Radiation	RadAvg		2.04	2.90	3.84	4.94	5.53	6.12	6.06	5.36	4.38	3.24	2.33	1.79
	RadStd		0.17	0.33	0.30	0.39	0.67	0.47	0.29	0.31	0.49	0.35	0.20	0.17

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (76 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+131	+76

Nomenclature: See separate page